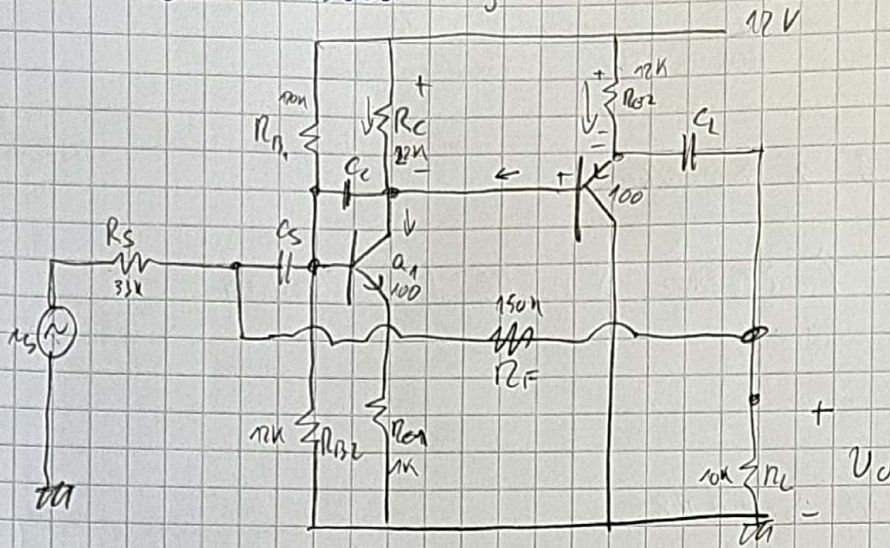




1) op point

$$V_{DD} = V_{GS} \frac{R_{D1}}{R_{D1} + R_{S1}} = 1.05V \quad R_{D1} = R_{D1} / R_{S1} = 10.9k\Omega$$

$$V_{D1} - R_{D1} I_{D1} - V_{GS} - R_{S1} I_{S1} = 0$$

$$I_{D1} = \frac{V_{D1} - V_{GS}}{R_{D1} + R_{S1}(1 + \beta_{F1})} = 3.485\mu A \quad \boxed{I_{C1} = 0.348mA}$$

$$-R_{C1} I_{C1} - V_{D2} + R_{D2} I_{D2} = 0 \quad -R_{C1}(I_{C1} - I_{B2}) - V_{D2} + R_{D2} I_{B2}(1 + \beta_{F2}) = 0$$

$$\begin{cases} I_{C1} = I_{C1} - I_{B2} \\ -R_{C1} I_{C1} + R_{C1} I_{B2} - V_{D2} + R_{D2} I_{B2}(1 + \beta_{F2}) = 0 \end{cases}$$

$$I_{B2} = \frac{R_{C1} I_{C1} + V_{D2}}{R_{C1} + R_{D2}(1 + \beta_{F2})} = 5.66\mu A \quad \boxed{I_{C2} = 0.566mA}$$

$$V_{C2} = -(V_{C1} - R_{C2} I_{C2}) = -5.23V$$

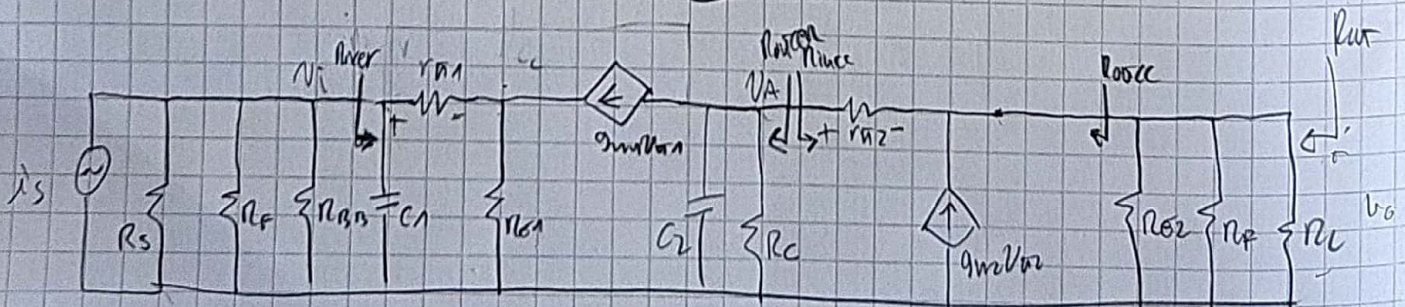
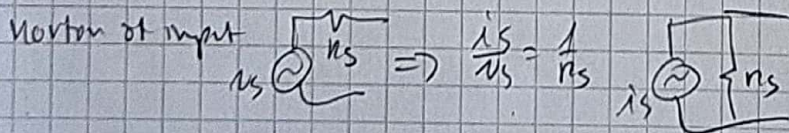
$$V_{C1} = V_{C1} - R_{C1} I_{C1} - R_{S1} I_{S1} = 3.99V$$

$$g_{m1} = 13.92mS^{-1} \quad g_{m2} = 22.56mS^{-1}$$

$$R_{in1} = 7.18k\Omega \quad R_{in2} = 4.43k\Omega$$

2) $A_v = V_o/V_s$ we have voltage sampling current mirror

$$\beta_{F1} \text{ return: } \beta = -\frac{1}{R_F} = -6.667 \cdot 10^{-6} S^{-1} \quad R_{F1} = R_{D1} = R_F = 150k\Omega$$



$$A_{ol} = \frac{v_o}{v_i} = \frac{v_o}{v_A} \cdot \frac{v_A}{v_i} \cdot \frac{v_i}{v_s}$$

$$\frac{v_o}{v_A} = \frac{(1+\beta_0)R_L}{R_{incc}} = 0,9544$$

$$\frac{v_A}{v_i} = \frac{-\beta_0(R_E \parallel R_{incc})}{R_{incc}} = -19,7$$

$$\frac{v_i}{v_s} = R_s \parallel R_E \parallel R_{in} \parallel R_{incc} = R_{in}^{ol} = 7,24 \text{ k}\Omega$$

$$A_{ol} = -141,3 \text{ k}\Omega \quad A_{VF} = \frac{A_{ol}}{1+\beta_0} = -72,7 \text{ k}\Omega$$

$$A_V = \frac{v_o}{v_s} = A_{VF} \cdot \frac{1}{R_s} = \boxed{-2,2}$$

$$3) R_{out}^o = R_i \parallel R_{outcc} = 24,8 \text{ k}\Omega$$

$$R_{out}^F = \frac{R_{out}^o}{1+\beta_0} = \boxed{127 \Omega}$$

$$4) C_c = 2 \text{ pF} \quad \text{split with miller, } G = A_{vcc} = -19,7$$

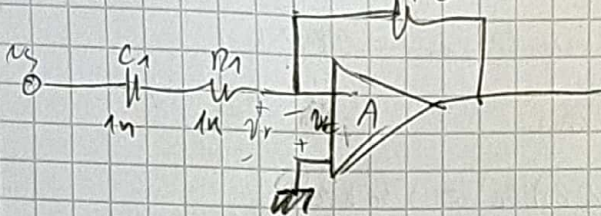
$$C_1 = 41,4 \text{ pF} \quad R_{c1}^o = R_{in}^{ol} = 7,24 \text{ k}\Omega$$

$$C_2 = 2,1 \text{ pF} \quad R_{c2}^o = R_E \parallel R_{incc} = 24,13 \text{ k}\Omega$$

$$f_H^o = \frac{1}{2\pi} \left(\frac{1}{C_1 R_{c1}^o + C_2 R_{c2}^o} \right) = 462,5 \text{ kHz}$$

$$f_H^F = f_H^o (1+\beta_0) = \boxed{898 \text{ kHz}}$$

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$$A(s) = \frac{10^5}{1 + s/10^2}$$

1) Übertragungsfunktion

$$A_v(s) = -\frac{Z_2}{Z_1} = -\frac{R_2}{R_1 + \frac{1}{sC_1}} = -\frac{R_2}{\frac{1 + sC_1R_1}{sC_1}} = -\frac{sC_1R_2}{1 + sC_1R_1}$$

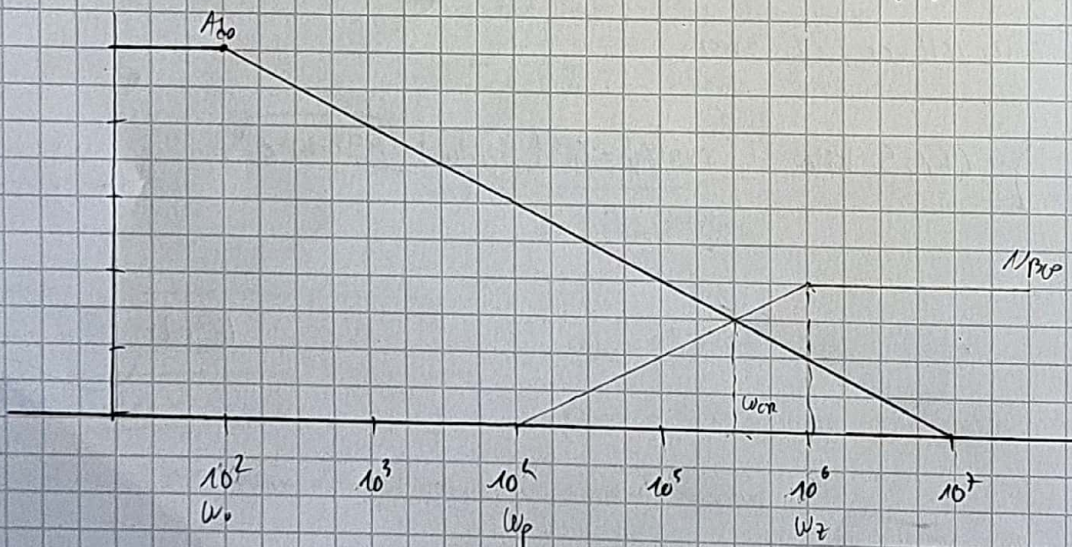
2) φ_m

$$T \approx -\frac{v_n}{v_e} = A(s) \cdot \frac{Z_1}{Z_2 + Z_1} = A(s) \cdot \frac{\frac{1 + sC_1R_1}{sC_1}}{R_2 + \frac{1 + sC_1R_1}{sC_1}} = A(s) \cdot \frac{1 + sC_1R_1}{1 + sC_1(R_1 + R_2)}$$

$$\omega_p = \frac{1}{C_1(R_1 + R_2)} = 9,9 \cdot 10^3 \text{ rad/s}$$

$$\omega_z = \frac{1}{C_1 R_1} = 10^6 \text{ rad/s}$$

$$\beta(s) = \begin{cases} \frac{1}{\beta(0)} = 1 = 0 \text{ dB} \\ \frac{1}{\beta(\infty)} = \frac{R_1 + R_2}{R_1} = 40 \text{ dB} \end{cases}$$



$$\begin{cases} A_0 \omega_0 = A \times \omega_{0z} \\ \frac{1}{\omega_p} = \frac{A}{\omega_{0z}} \end{cases} \Rightarrow \omega_{0z} = \sqrt{A_0 \omega_0 \omega_p} = 3,14 \cdot 10^5 \text{ rad/s}$$

$$\varphi_m = 180 - \arctan\left(\frac{\omega_{0z}}{\omega_0}\right) - \arctan\left(\frac{\omega_{0z}}{\omega_p}\right) + \arctan\left(\frac{\omega_{0z}}{\omega_z}\right) = \boxed{19,3^\circ} \sim 10^\circ$$

3) R_1 to have $\varphi_m = 45^\circ$

cosy: put $\omega_z = \omega_{cn}$

more current (not perfect)

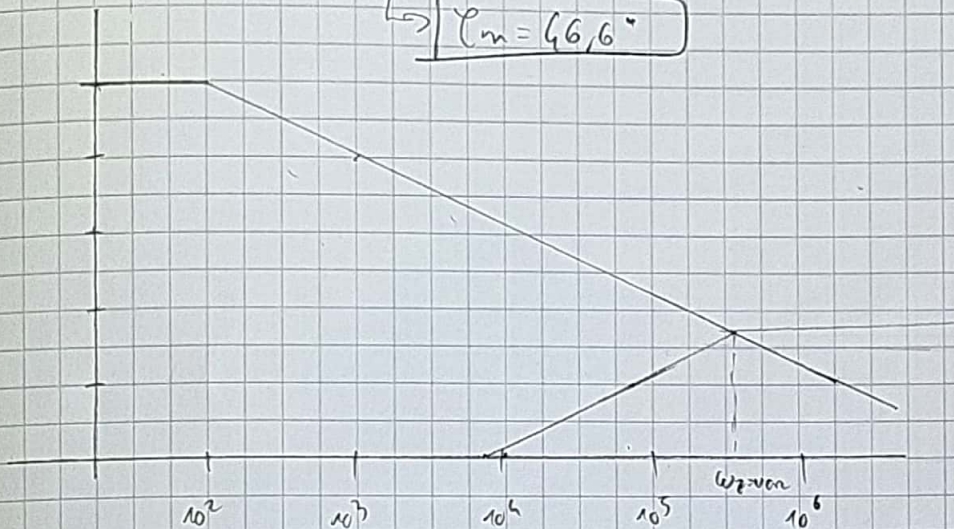
$$19,3 + \arctan\left(\frac{\omega_{cn}}{\omega_z}\right) = 45 \rightarrow \frac{\omega_{cn}}{\omega_z} = 0,68$$

$$\omega_z = \omega_{cn} \rightarrow R_1 = \frac{1}{\omega_{cn} C_1} = \boxed{3178 \Omega}$$

$$\omega_{zc} = \omega_{cr}$$

$$\omega_{pc} = 9,6 \cdot 10^3 \text{ rad/s} \approx \omega_p$$

$$\hookrightarrow \boxed{\varphi_m = 46,6^\circ}$$



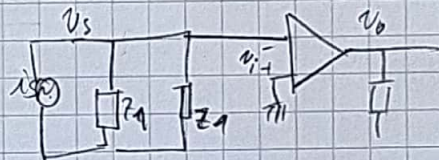
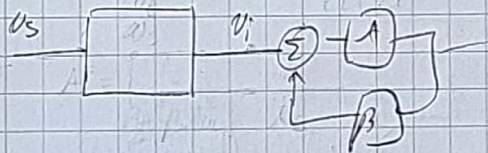
problem

$$A_{ol} \omega_0 = 1/p_{ol} \omega_{cn} \quad \omega_{cn} = A_{ol} \omega_0 \frac{R_1}{R_1 + R_2} \quad 180 - \arctan\left(\frac{A_{ol} \omega_0 \frac{R_1}{R_1 + R_2}}{1}\right) - \arctan\left(\frac{A_{ol} \omega_0 \frac{R_1}{R_1 + R_2}}{1}\right) = 45^\circ$$

1) gain of 10^3 rad/s

$$\boxed{R_1 = 3106 \Omega}$$

$$T = -\frac{v_o}{v_i} = A(s) \cdot \frac{z_1}{z_1 + z_2}$$



$$A_F = \frac{v_o}{v_i} = \frac{A(s)}{1 + A(s) \frac{z_1}{z_1 + z_2}}$$

$$A_F = \frac{v_o}{v_i} = \frac{v_o}{v_i} \cdot \frac{1}{1 + A(s) \frac{z_1}{z_1 + z_2}}$$

$$A_F = \frac{v_o}{v_i} = \frac{v_o}{v_i} \cdot \frac{v_i}{v_s} = \frac{A(s)}{1 + A(s) \frac{z_1}{z_1 + z_2}} \cdot \frac{z_2}{z_1 + z_2} = \frac{z_2}{z_1 + z_2 + z_1} = \boxed{0,96}$$

$$R_1 = 3178$$

$$z_1(10^3) = R_1 + \frac{1}{j10^3 C_1} = \boxed{3178 \Omega}$$

$$z_2 = R_2 = 10000 \Omega$$

interference term

$$A_{ol} = -A_d(z_1/R_2) \omega/\omega_s$$

$$A_F = \frac{-A_d(z_1/R_2)}{1 + A_d(s)(z_1/R_2)} \cdot \frac{1}{z_1} = \frac{v_o}{v_s}$$